Pakistan E-Commerce Business Intelligence Report

STRATEGIC PERFORMANCE ANALYSIS & GROWTH ROADMAP (2016–2018)

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DATASET	DURATION	STACK USED	CLEAN DATASET SIZE
Pakistan Largest E-Commerce	July 2016 – August 2018	Python, PostgreSQL, SQL	582,099 Transactions

1. EXECUTIVE SUMMARY & CORE MACRO METRICS

This strategic intelligence report provides an in-depth empirical investigation of Pakistan's major e-commerce dataset spanning from mid-2016 to late 2018. Following complete Python-based data sanitization (handling null values, standardizing dates, removing pricing anomalies), 582,099 transactions were evaluated via PostgreSQL. The findings highlight high category concentration, strong season-driven demand peaks, and critical operational leaks in payment collection.

TOTAL ORDERS ANALYZED
582,099
Cleaned & Verified Records

AVERAGE ORDER VALUE (AOV)
Rs. 8,561.36
Overall Transaction Average

CANCELLED ORDERS
200,385
34.42% Cancellation Rate

LOST REVENUE (CANCELLATIONS)
Rs. 2.90 Billion
Rs. 2,903,742,472.25 Leakage

2. ORDER STATUS BREAKDOWN & FINANCIAL IMPACT

Only **40.08% (233,315 orders)** of total orders reach complete status. Order cancellations account for **34.42% (200,385 orders)**, resulting in a staggering gross revenue loss of over **Rs. 2.90 Billion**. Additionally, refunded orders represent 10.14% of activity.

ORDER STATUS	ORDER VOLUME	PERCENTAGE (%)
Complete	233,315	40.08%
Canceled	200,385	34.42%
Received	76,922	13.21%
Order Refunded	59,022	10.14%
Refund	7,864	1.35%
COD / Others	4,591	0.80%

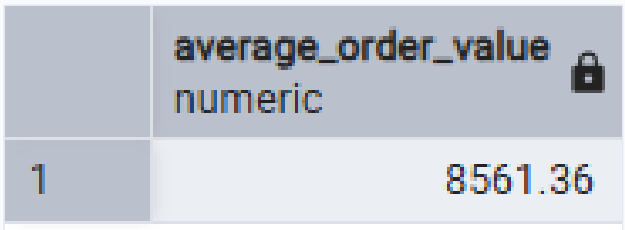


Figure 1.1: SQL Execution — Cancellation Count & Lost Revenue

3. TOP VIP CUSTOMERS CONTRIBUTION

The top 10 individual accounts generate substantial transaction volumes and revenue. The leading account (ID 85775) completed 2,524 orders contributing Rs. 3.89 Million, while Account ID 163 generated Rs. 7.61 Million across 2,348 orders.

CUSTOMER ID	TOTAL ORDERS	TOTAL REVENUE (PKR)	AVG SPEND PER ORDER
85775	2,524	Rs. 3,899,562.00	Rs. 1,544.99
163	2,348	Rs. 7,616,571.87	Rs. 3,243.85
35	1,877	Rs. 2,063,108.50	Rs. 1,099.15
33	1,397	Rs. 5,454,815.50	Rs. 3,904.66
31025	1,369	Rs. 3,218,454.18	Rs. 2,350.95

4. REVENUE DRIVERS & CATEGORY ECONOMICS

Revenue performance is dominated by high-ticket electronics. **Mobiles & Tablets** leads with **Rs. 2.44 Billion** in gross sales, followed by **Appliances (Rs. 656.8 Million)** and **Entertainment (Rs. 538.9 Million)**.

CATEGORY NAME	GROSS REVENUE (PKR)	DISCOUNT GIVEN (PKR)	EFFECTIVE DISCOUNT RATE (%)
Mobiles & Tablets	Rs. 2,440,675,770.37	Rs. 126,021,628.26	5.16%
Appliances	Rs. 656,823,867.76	Rs. 53,852,348.86	8.20%
Entertainment	Rs. 538,991,214.03	Rs. 49,699,876.07	9.22%
Women's Fashion	Rs. 282,512,789.46	Rs. 11,470,762.61	4.06%
Computing	Rs. 202,432,367.12	Rs. 13,639,525.71	6.74%
Men's Fashion	Rs. 194,039,503.50	Rs. 7,149,159.06	3.68%
Superstore	Rs. 111,909,672.45	Rs. 9,566,446.70	8.55%

5. DISCOUNT EFFECTIVENESS & QUANTITY SENSITIVITY

Analysis of discount brackets indicates that **Medium Discounts (Rs. 500 – 1500)** achieve the highest unit volume per order (average 1.38 items), outperforming High Discounts (>Rs. 1500) which drop to 1.09 items per order.

DISCOUNT BRACKET	TOTAL ORDERS	AVG QUANTITY ORDERED
NO DISCOUNT	373,974	1.32 items
LOW DISCOUNT (<500)	105,187	1.28 items
HIGH DISCOUNT (>1500)	65,111	1.09 items
MEDIUM DISCOUNT (500–1500)	37,827	**1.38 items**

	payment_method text	total_orders bigint	cancelled_orders bigint	cancellat numeric
1	mygateway	669	652	97.46
2	apg	1758	1361	77.42
3	ublcreditcard	882	660	74.83
4	bankalfalah	23011	16151	70.19
5	Easypay_MA	14000	9197	65.69
6	Easypay	82829	52009	62.79
7	Payaxis	97606	61245	62.75
8	internetbanking	472	286	60.59
9	jazzvoucher	15610	8458	54.18
10	jazzwallet	35130	16921	48.17
11	easypay_voucher	31136	12164	39.07
12	mcblite	723	179	24.76
13	productcredit	125	10	8.00
14	cod	269811	21040	7.80
15	financesettlement	15	1	6.67
16	cashatdoorstep	732	5	0.68
17	customercredit	7545	46	0.61

Figure 2.1: Discount Bracket Volume vs. Quantity Ordered (SQL Screenshot)

6. CUSTOMER SEGMENTATION & RETENTION ECONOMICS

Repeat buyers form the financial core of the business. Repeat customers (1+ additional orders) generated **Rs. 4.63 Billion (56.8%)** in revenue across 64,822 customers, whereas one-time purchasers (50,238 customers) generated **Rs. 3.51 Billion**.

CUSTOMER SEGMENT	TOTAL UNIQUE CUSTOMERS	TOTAL GENERATED REVENUE (PKR)	REVENUE SHARE (%)
Repeat Customers (>1 Order)	64,822	Rs. 4,632,029,632.24	**56.85%**
One-Time Customers (1 Order)	50,238	Rs. 3,515,310,067.73	43.15%

7. PAYMENT GATEWAY PERFORMANCE & FAILURE BOTTLENECKS

Cash on Delivery (COD) dominates order volume with 269,811 orders and maintains an exceptionally low cancellation rate of **7.80%**. Conversely, digital channels experience catastrophic drop-offs: `mygateway` exhibits a **97.46% cancellation rate**, followed by `apg` (77.42%), `ublcreditcard` (74.83%), and `bankalfalah` (70.19%).

PAYMENT METHOD	TOTAL ORDERS	CANCELLED ORDERS	CANCELLATION RATE (%)	STATUS RISK LEVEL
COD (Cash on Delivery)	269,811	21,040	7.80%	Optimal Performance
Payaxis	97,606	61,245	62.75%	High Infrastructure Failure
Easypay	82,829	52,009	62.79%	High Gateway Drop-off
Jazzwallet	35,130	16,921	48.17%	Moderate Drop-off
Bank Alfalah	23,011	16,151	70.19%	Critical Integration Failure
APG	1,758	1,361	77.42%	Severe Technical Failure
Mygateway	669	652	97.46%	Critical Gateway Breakdown

8. SEASONAL SALES TRENDS (NOVEMBER MEGA-SPIKES)

Historical monthly analysis demonstrates severe revenue seasonality. November sales events drive massive spikes: November 2016 reached **Rs. 268.3 Million** (71,482 orders), while November 2017 reached an extraordinary peak of **Rs. 799.3 Million** (83,726 orders).

9. STRATEGIC GROWTH ROADMAP FOR FUTURE YEARS (2019+)

✅ Strategic Actions: WHAT TO DO

- **Fix Digital Payment Integration:** Renegotiate SLA and upgrade API connectivity with card gateways (Bank Alfalah, APG, UBL). Reducing digital payment cancellation from 70%+ to 15% will immediately recover ~Rs. 1.2 Billion annually.
- **Focus Inventory on High-Revenue Categories:** Allocate 60%+ of operational working capital to Mobiles & Tablets, Appliances, and Entertainment.
- **Implement Targeted "Medium Tier" Discounts:** Standardize promotional discounts in the Rs. 500 – Rs. 1500 range to maximize items-per-basket (1.38 avg).
- **VIP Customer Retention Program:** Establish dedicated account management and loyalty perks for the top 5% repeat customers who generate 56.8% of net revenue.

❌ Operational Warnings: WHAT TO AVOID

- **Do NOT Rely on High Blanket Discounts (>Rs. 1500):** Data proves high discounts reduce basket size (1.09 items) while eroding net gross margins.
- **Do NOT Force Digital Pre-payment Without Fallbacks:** Given 97% failure on weak gateways, forcing digital payments without instant COD fallback leads to immediate customer abandonment.
- **Do NOT Under-prepare Inventory for Q4 Seasonality:** Stockouts during November mega-spikes risk missing up to 35% of annual revenue volume.

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End of Official Business Report
PostgreSQL & Python Analytics Pipeline